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Syntactic Parsing in Post-Stroke Aphasia: The Role of Cerebellar-Cerebral Functional Connectivity

Introduction: The right posterior cerebellum (Crus I/II) is increasingly recognized as contributing to language processing (Stoodley & Schmahmann, 2009), including sentence-level integration and syntactic computation (Xu et al., 2005; Nakatani et al., 2022). In post-stroke aphasia, cerebro-cerebellar functional connectivity (FC) has been linked to language outcomes such as naming (Stilling et al., 2025), suggesting that cerebellar-cortical interactions may support recovery. However, its role in supporting sentence comprehension, particularly under varying syntactic demands, remains largely unknown. This study tests whether resting-state FC between right Crus I/II and cortical language regions predicts performance in various conditions of English grammaticality judgment and syntactic comprehension tasks.

Methods: 40 individuals with chronic left-hemisphere stroke and aphasia (mean age = 57.2; education = 15 years; mean Aphasia Quotient [AQ] = 80.1) completed resting state fMRI and two English behavioral tasks. In a grammaticality judgement task, participants read sentences aloud and indicated whether they were well-formed. In a syntactic comprehension sentence-picture matching task, participants had to match pictures to sentences of varying syntactic complexity. Sentences were presented either auditorily or visually, and the two modalities were administered separately. Actives and subject-extracted clefts were classified as 'simple', while object-extracted clefts and object relative clauses were considered 'complex'. Effects of trial type (grammatical vs. ungrammatical; simple vs. complex) and modality (auditory vs. visual) were assessed using Wilcoxon signed-rank tests. FC was computed between right cerebellar Crus I/II and 38 bilateral cortical language regions. FC values of significant connections were extracted for prediction analyses. First, we used LASSO regression to select the FC connections that best predicted accuracy for each task. Retained connections were then entered into linear regression models that also included covariates (age, education, lesion volume, and AQ). Each full model (FC + covariates) was compared to a covariate-only model using ANOVA. Analysis was repeated with a negative control region (left cerebellar lobules 4/5).

Results: Participants showed higher accuracy for grammatical vs. ungrammatical sentences ($p<0.001$) and for simple vs. complex syntactic structures ($p<0.001$). There were no differences in accuracy between auditory and visual trials, so modalities were averaged for subsequent analyses. LASSO identified no Crus I/II-seeded FC predictors of performance on grammaticality judgement. Several connections were retained for the syntactic comprehension task (Figure 1), with stronger effects observed for complex sentences. Including these retained FC predictors in linear regressions increased variance explained by 19% for simple trials and 49% for complex trials relative to covariate-only models (both $p < .001$; Table 1). For the control seed, 3 intra-cerebellar connections were identified as important in the syntax task (Figure 1), but model improvement was much lower than Crus I/II-seeded FC (Table 1).

Conclusion: These findings suggest that cerebellar-cerebral FC connections contribute to the computational demands of syntactic processing in aphasia. No predictors emerged for grammaticality judgment, a task in which most violations could be detected locally, without constructing full hierarchical syntactic representations. In contrast, cerebellar-cortical FC substantially improved prediction of syntactic comprehension accuracy, particularly for complex, noncanonical structures that require long-distance dependency resolution and hierarchical integration.

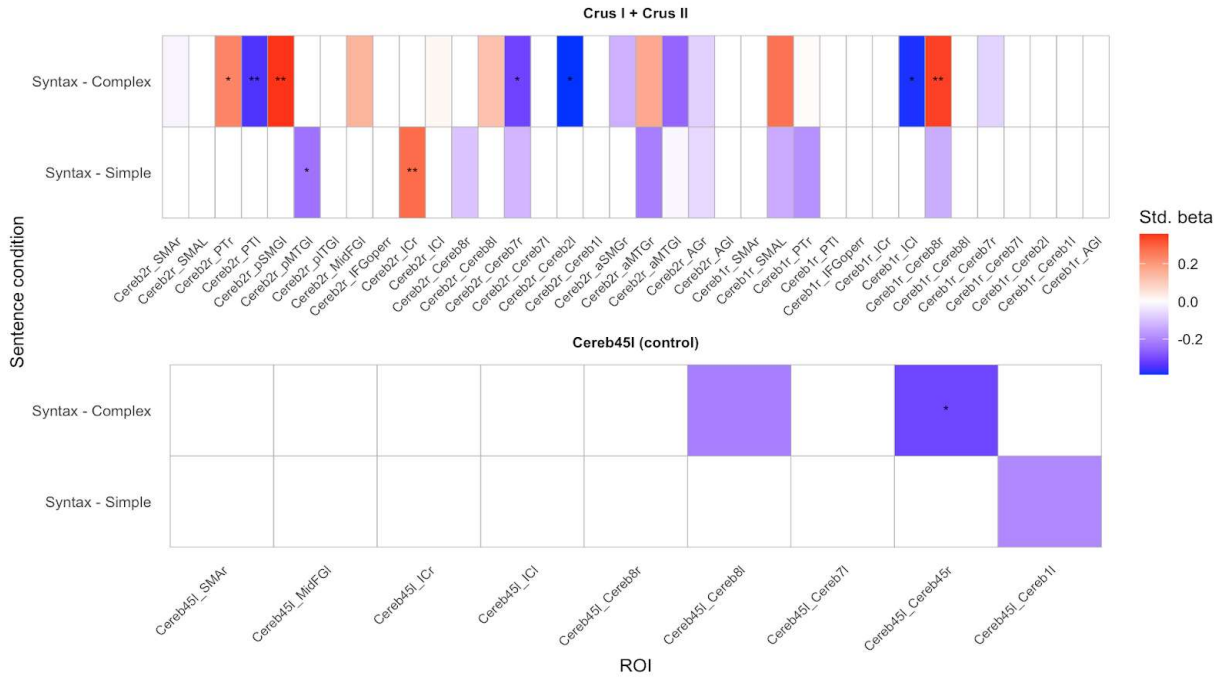

Figure 1: Significant ROIs per outcome measure. Colored cells indicate ROIs selected by LASSO, including significance of predictors in final models (*: p<0.05; **: p<0.01; ***: p<0.001).

Table 1: Linear regression performance for models with covariates only (base), with FC added (final), and the improvement in variance explained (Δ).

Seed	Outcome	Base Adj. R^2	Final Adj. R^2	ΔAdj. R^2	Anova p
Crus I/II R	Grammatical	0.64	0.64	0	NA
	Ungrammatical	0.38	0.38	0	NA
	Simple	0.62	0.81	+0.19	0.001
	Complex	0.30	0.79	+0.49	<0.001
Lobule 4/5 L (control)	Grammatical	0.64	0.64	0	NA
	Ungrammatical	0.38	0.38	0	NA
	Simple	0.62	0.65	+0.02	0.09
	Complex	0.30	0.44	+0.14	<0.01

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